



Robust 3D Vision Modeling

Under Diverse Adverse Environmental Conditions

종합설계 프로젝트2 - 3팀 발표자: 김규도

KYUNGPOOK NATIONAL UNIVERSITY

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& Operational Process**
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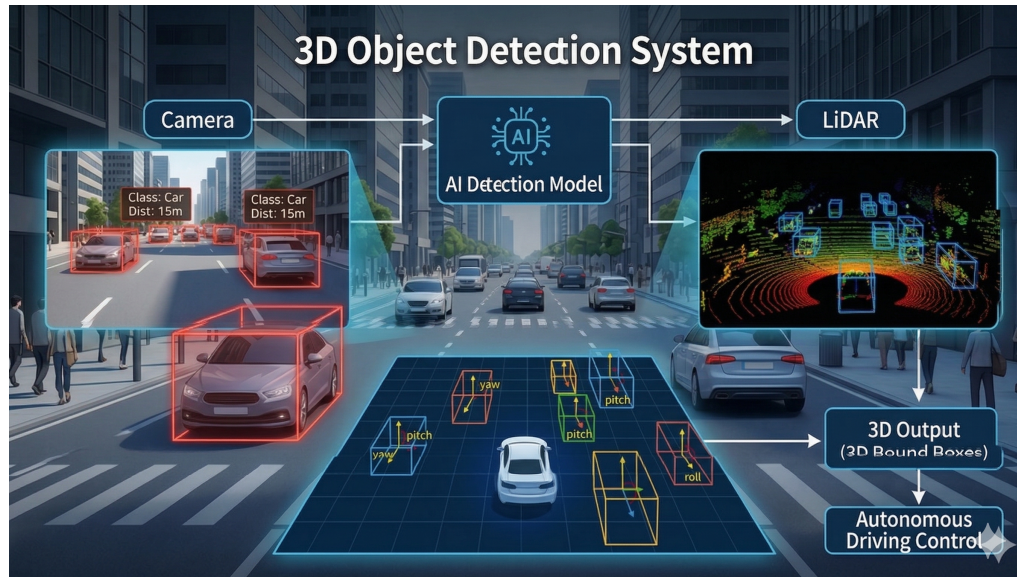
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Introduction



Background and Objectives

0. Introduction



A.

3DOD_SNOW

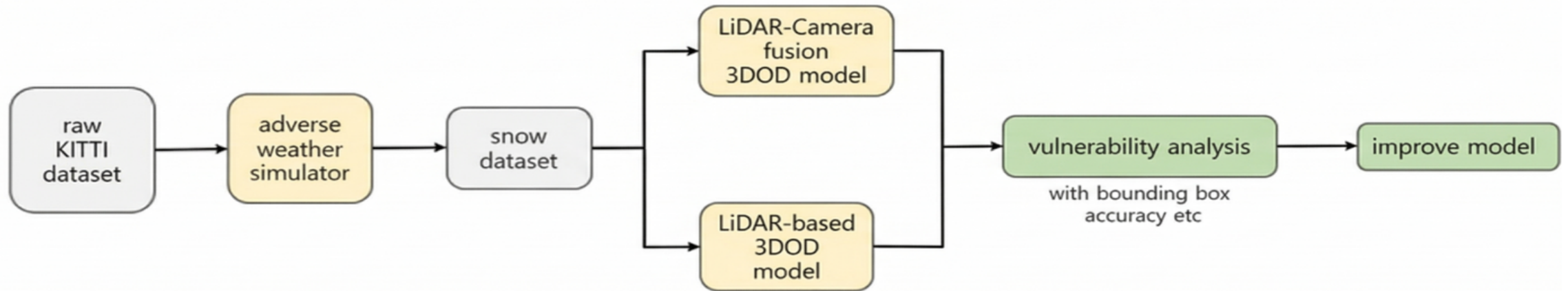
담당 연구자: 이상욱님



1. Architecture & Process

A.

3DOD_SNOW



- Exclusion of camera-only models
- Metric: Localization Accuracy (mAP)

■ 2. Progress and Challenges

A.

~~3DOD_SNOW~~

- Dataset: **kitti dataset**
- Weather Simulator: **3D_Corruptions_AD**
- 3DOD Model:
 - LiDAR : **PointPillars/PointRCNN/Voxel RCNN**
 - LiDAR+image fusion : **MVMM**
- GPU: RTX 5060 Ti
- Software: Python
- OS: Ubuntu

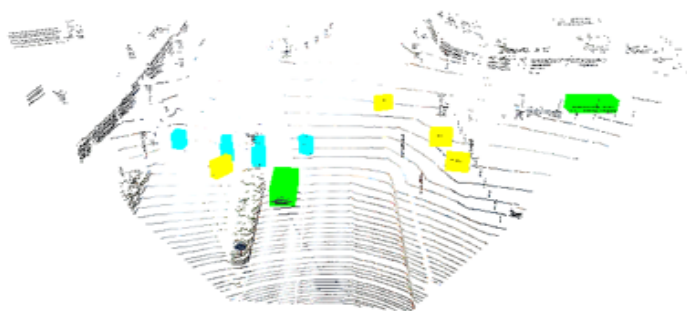
2. Progress and Challenges

A.

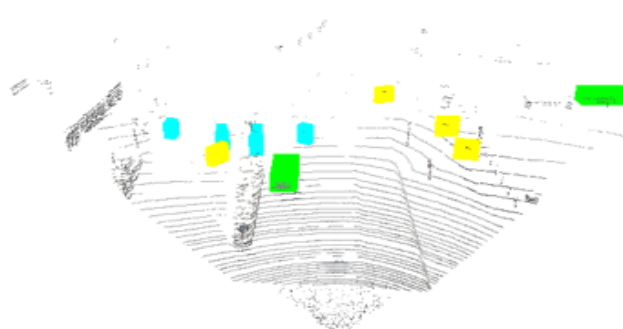
3DOD_SNOW

PointPillars : 006813

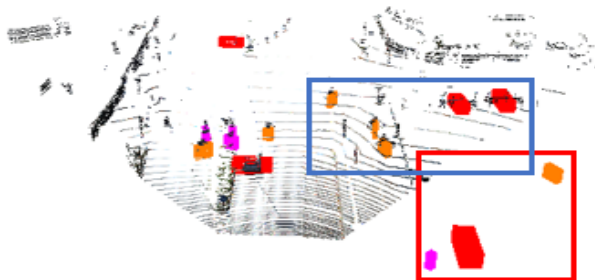
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



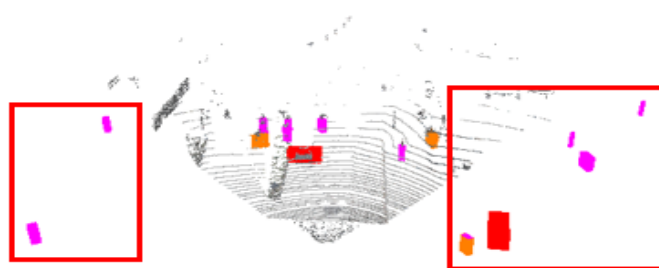
kitti_GT



Snow kitti_GT



Kitti pred



Snow kitti pred

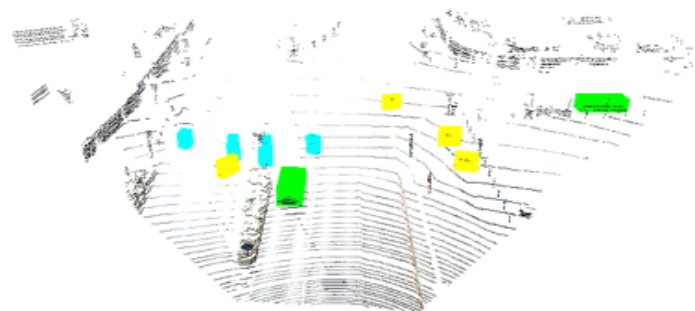
2. Progress and Challenges

A.

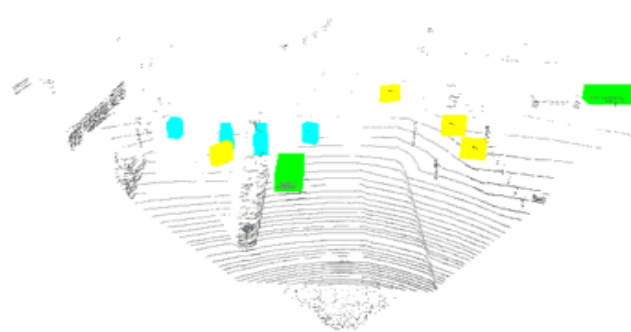
3DOD_SNOW

PointRCNN : 006813

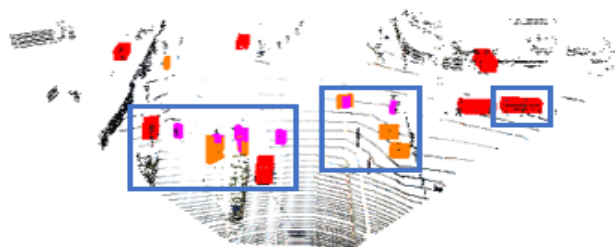
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GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



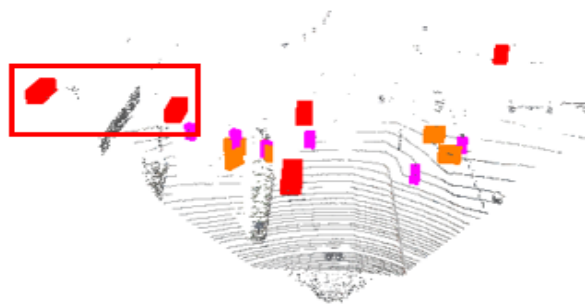
kitti_GT



Snow kitti_GT



Kitti_pred



Snow_kitti_pred

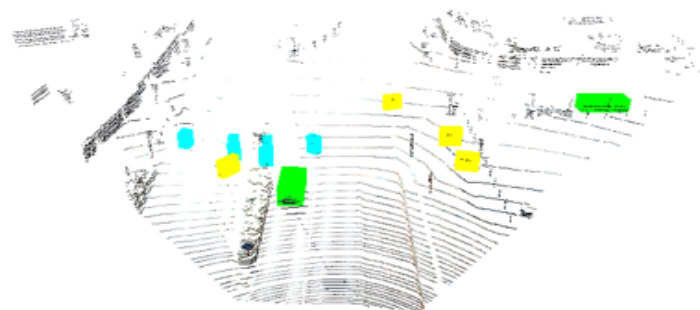
2. Progress and Challenges

A.

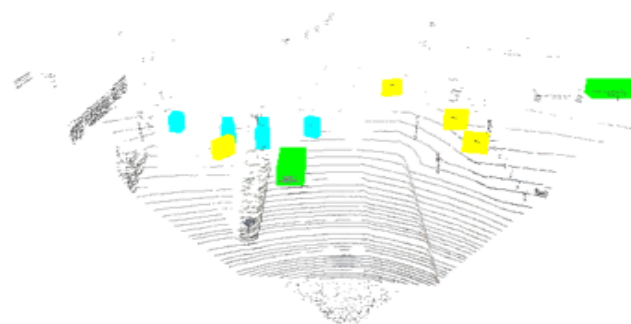
3DOD_SNOW

VoxelRCNN : 006813

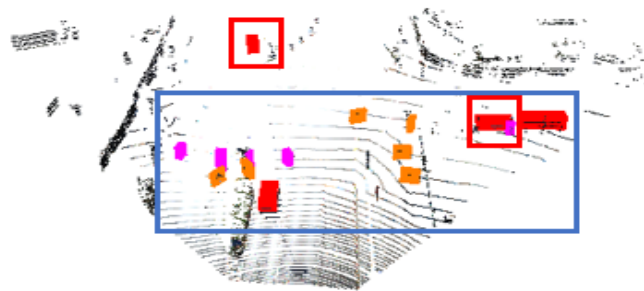
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



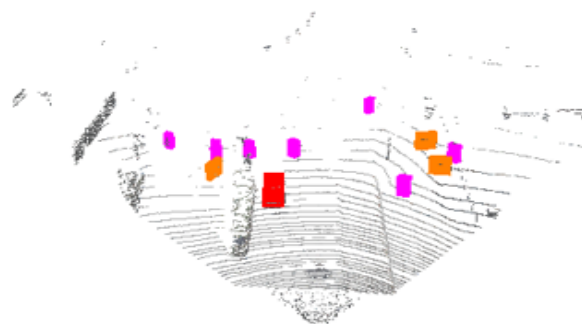
kitti_GT



Snow kitti_GT



Kitti pred



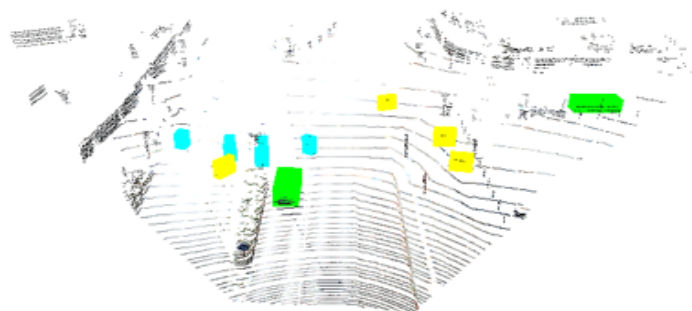
Snow pred

2. Progress and Challenges

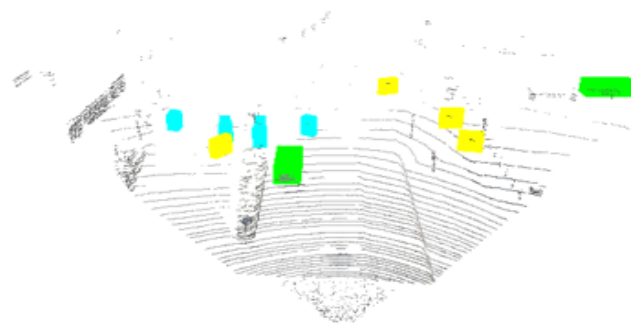
A.

3DOD_SNOW

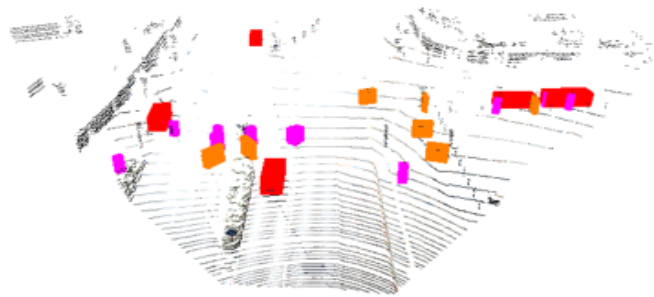
Mvmm : 006813



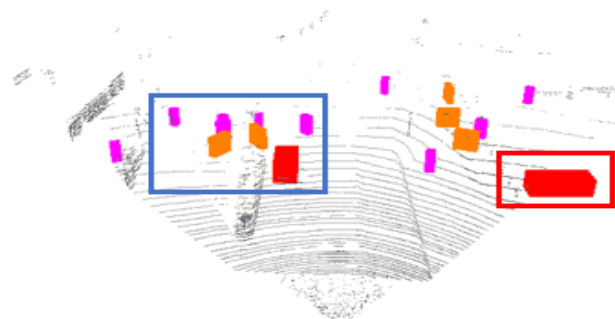
kitti_GT



Snow kitti_GT



Kitti_pred



Snow_pred

	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

- Performance degradation at long range from LiDAR

2. Progress and Challenges

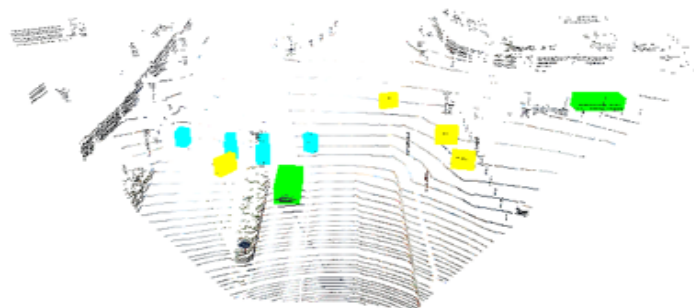
A.

3DOD_SNOW

Scene_num : 006813

모델 별 결과 비교 – raw kitti

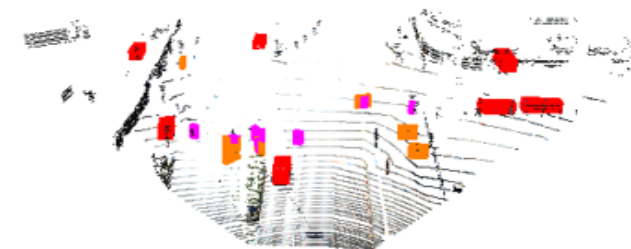
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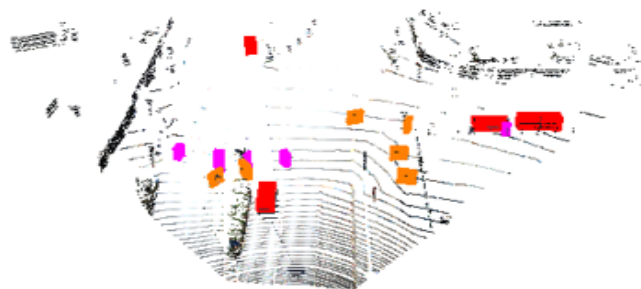
kitti_GT



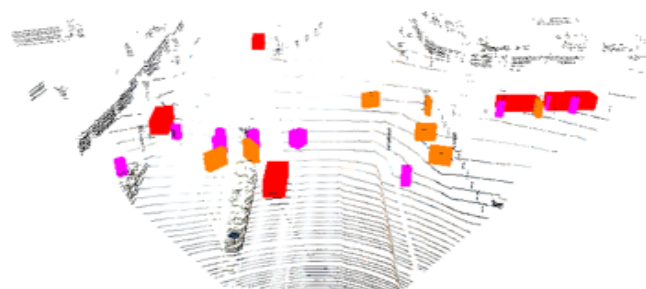
PointPillars



PointRCNN



VoxelRCNN



mvmm

- The best-in-class model:
VoxelRCNN, MVMM

2. Progress and Challenges

A.

3DOD_SNOW

Repo	Metric	Pedestrian AP@0.5	Cyclist AP@0.5	Car AP@0.7	GPU
VoxelRCNN_repo	3D-Bbox	-	-	-	-
VoxelRCNN_kitti	3D-Bbox	66.5348, 60.2782, 55.6542	85.9466, 72.1112, 68.8626	89.6149, 83.8685, 78.8079	5060ti x1
VoxelRCNN_snow	3D-Bbox	66.6909, 59.3986, 55.9992	83.5091, 59.1099, 55.1060	74.8540, 56.2723, 50.7544	5060ti x1
VoxelRCNN_repo	BEV	-	-	-	
VoxelRCNN_kitti	BEV	67.7843, 63.0610, 59.7228	92.3742, 74.5277, 72.2761	90.2165, 88.1085, 87.6400	
VoxelRCNN_snow	BEV	70.8167, 64.0834, 58.8447	87.0567, 60.0769, 57.7704	79.2096, 66.7585, 60.1988	
VoxelRCNN_repo	2D-BBox	-	-	-	
VoxelRCNN_kitti	2D-BBox	75.4673, 71.4710, 67.2322	94.7069, 82.7996, 80.4283	97.5992, 89.6159, 89.2497	
VoxelRCNN_snow	2D-BBox	75.7812, 68.8897, 66.4404	88.2552, 64.6207, 60.7499	89.7385, 71.0313, 70.0774	
VoxelRCNN_repo	AOS	-	-	-	
VoxelRCNN_kitti	AOS	70.46, 66.24, 62.10	94.51, 82.32, 79.90	97.55, 89.50, 89.06	
VoxelRCNN_snow	AOS	70.80, 64.04, 61.58	87.88, 64.09, 60.24	89.37, 70.42, 69.21	

- Noteworthy metrics : **Metric of Pedestrian**

B. 3DOD_NIGHT

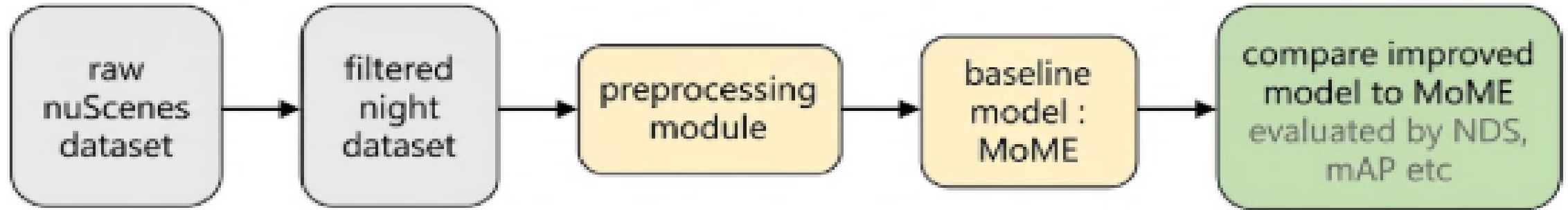
담당 연구자: 최경진님



1. Architecture & Process

B.

3DOD_NIGHT



- **Preprocessing module**
 - Illumination Compensation
 - Shadow Noise Reduction



- **Alleviate problems in low-light conditions**

■ 2. Progress and Challenges

B.

~~3DOD_NIGHT~~

- Dataset: **nuScenes dataset**
- Weather Simulator: **X (Real-world night scene)**
- 3DOD Model: **MoME**

- GPU: RTX 2080 Ti
- Software: Python
- OS: Ubuntu

2. Progress and Challenges

B.

3DOD_NIGHT

MoME		AWARDistill		ContextualFusion	
항목	내용	항목	내용	항목	내용
학회	CVPR 2025	학회/저널	Expert Systems With Applications, 2024	학회/저널	IEEE 2024
입력	LiDAR + Camera	입력	LiDAR + Camera	입력	LiDAR + Camera
평가 데이터셋	nuScenes, nuScenes-R, nuScenes-C	평가 데이터셋	KITTI, KITTI-C, KITTI-M, nuScenes, nuScenes-C, WOD-DA	평가 데이터셋	nuScenes, AdverseOp3D (CARLA 기반 자체 합성 데이터셋)
방법론	Mixture of Experts + Adaptive Query Router	방법론	Staged Knowledge Distillation (clean teacher → adverse student)	방법론	환경 맥락 파악 후 센서 융합 전략 조절
타겟 조건	센서 고장 (LiDAR/Camera drop 등)	타겟 조건	Fog, Rain, Snow, Strong Light	타겟 조건	Rain, Night
github	https://github.com/konyul/MoME (weight 공개 O)	github	공개된 링크 없음	github	https://github.com/ssuralcmu/ContextualFusion (pretrained model 공개o)

Improving Robustness of LiDAR		TransFusion	
항목	내용	항목	내용
학회/저널	arXiv 2024	학회/저널	CVPR 2022
입력	LiDAR + Camera	입력	LiDAR + Camera
평가 데이터셋	KITTI, nuScenes, nuScenes-c	평가 데이터셋	nuScenes, Waymo Open Dataset
방법론	Flexibly Weighted Feature Fusion (환경 맥락에 따라 LiDAR와 Camera 특징의 반영 비율을 동적으로 조절)	방법론	TransFusion (3D Query를 통해 이미지 특징을 적응적으로 찾아내어 융합)
타겟 조건	Fog, Rain, Snow, Sunlight	타겟 조건	야간,저조도 등 이미지 화질 저하 환경
github	공개 링크 없음	github	https://github.com/XuyangBai/TransFusion

2. Progress and Challenges

B.

3DOD_NIGHT

• nuScenes 데이터셋 계열 (metric : mAP)

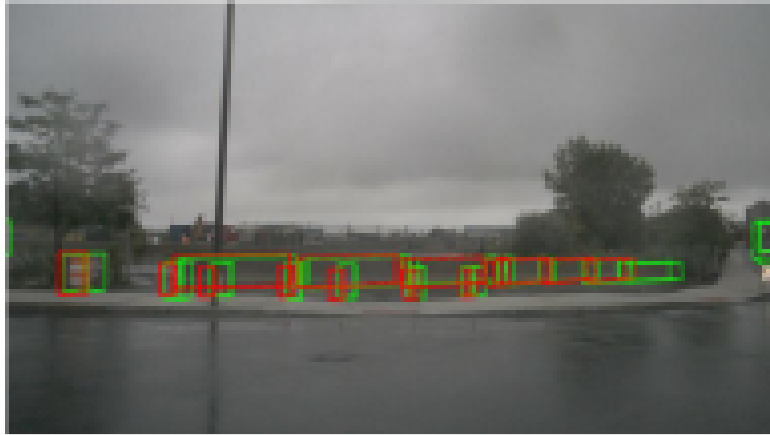
Dataset (Metric)	Model	Modality	Clean	Rain	Night	Fog	Snow
nuScenes (Real)	MoME	LC	71.20	72.2	43.0	-	-
	ContextualFusion (Independent)	LC	67.20	68.9	73.0	-	-
	TransFusion	LC	65.70	-	55.2	-	-
nuScenes-C (Synthetic)	MoME	LC	71.20	-	-	67.90	63.50
	AWARDistill	LC	61.30	66.93	-	60.11	66.03
	Transfusion	LC	65.70	-	--	53.70	63.30

- **High Potential** for Improvement
- Elevate **the overall reliability of the strongest base line**

2. Progress and Challenges

B.
3DOD_NIGHT

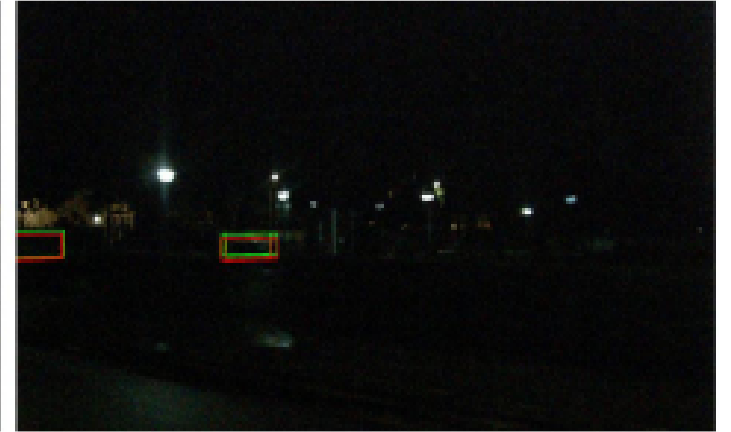
CAM_FRONT_LEFT



CAM_FRONT



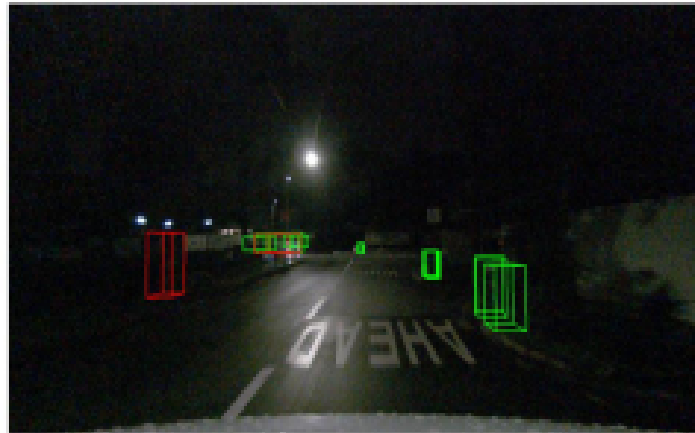
CAM_FRONT_RIGHT



CAM_BACK_LEFT



CAM_BACK



CAM_BACK_RIGHT



2. Progress and Challenges

B.

3DOD_NIGHT

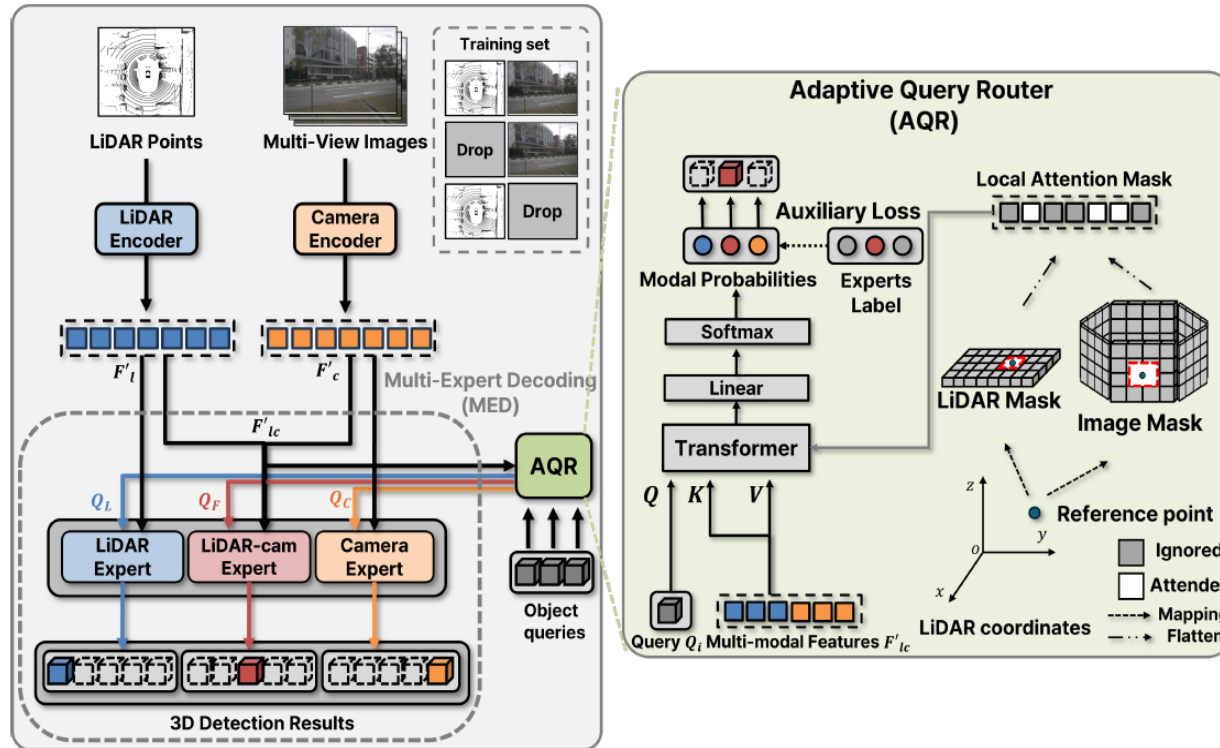
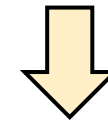


Figure 3. **Overall structure of MoME.** MoME utilizes three expert decoders, each specialized for processing LiDAR, camera, or LiDAR-camera features. The AQR dynamically assigns each object query to the most suitable expert decoder in an adaptive manner. The operation of AQR relies on local features filtered through the Local Attention Mask.

- architectural improvement to MoME's **weight distribution**
- **Preprocessing module** (Improving image quality)
 - limited novelty



- **Low-light feature extractor**
- **Knowledge Distillation (KD)** leveraging **clean-weather data**

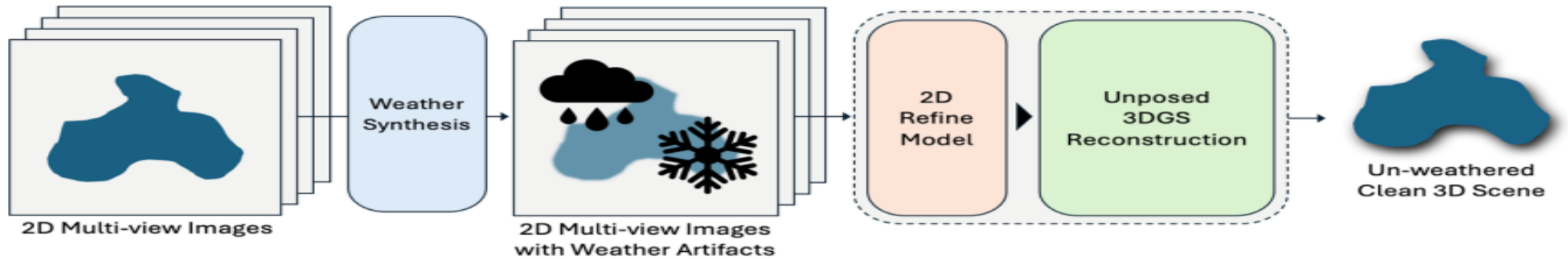
C. NVS_3DGS

담당 연구자: 이윤호님



1. Architecture & Process

C. NVS_3DGS



- **Structure from Motion (SfM)**
- **Posed/Unposed:** With/Without SfM processing
- **Metric: PSNR (Peak Signal-to-Noise Ratio)**
 - + Camera pose estimation accuracy of LongSplat

■ 2. Progress and Challenges

C. NVS_3DGS

- Dataset: **WeatherGS dataset + Free Dataset**
- Weather Simulator: **WeatherEdit**
- 3DGS Baseline Model: **Vanilla 3DGS + LongSplat(Unposed)**
- 2D Refinement Model: **MWFormer**

- GPU: RTX 2080 Ti
- Software: Python
- OS: Ubuntu

2. Progress and Challenges

C. NVS_3DGS



2. Progress and Challenges

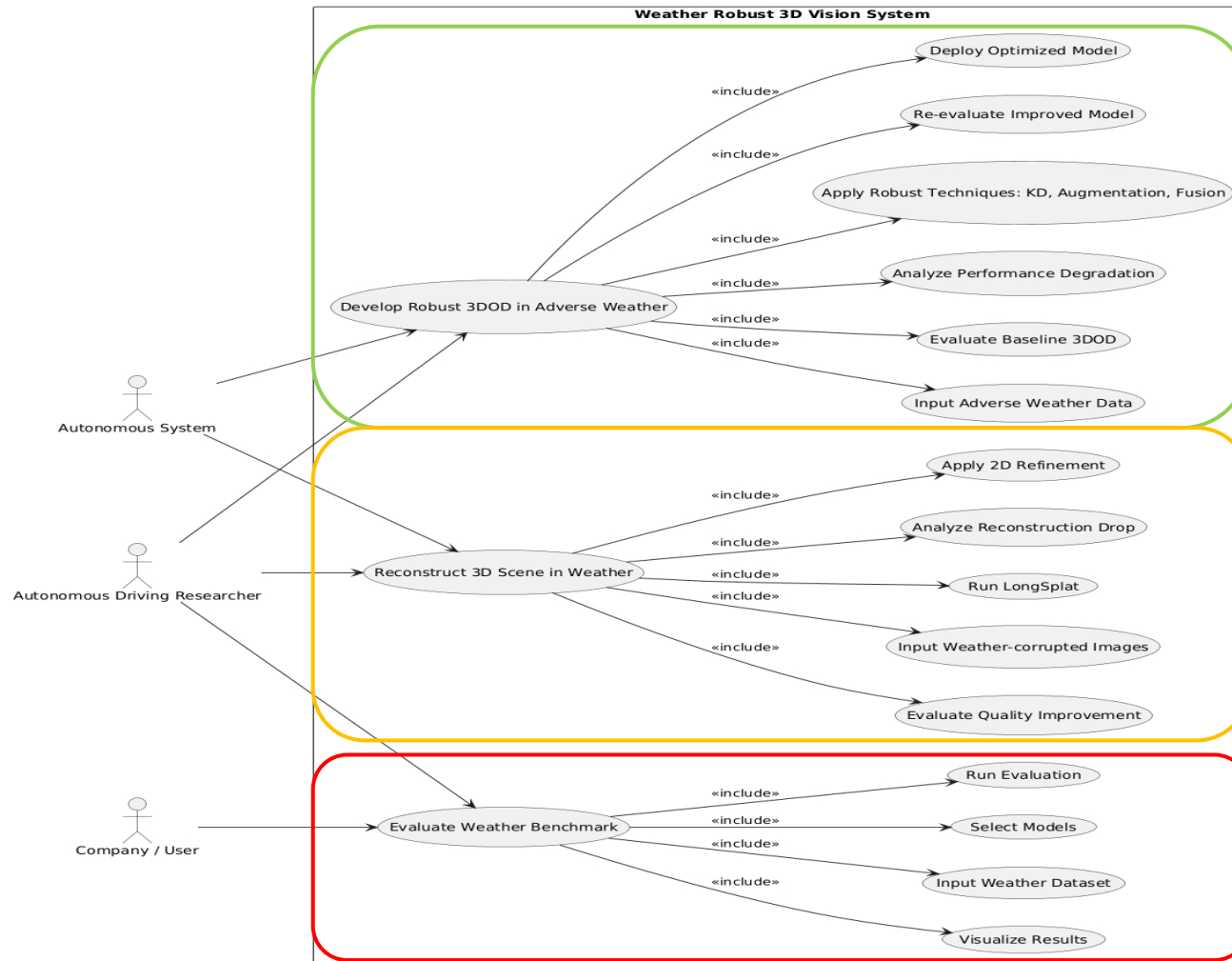
C. NVS_3DGS



	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
grass	24.057	0.741	0.197
pillar	<u>26.685</u>	<u>0.826</u>	<u>0.163</u>
road	23.934	0.735	0.254
sky	21.208	0.726	0.231
stair	27.715	0.852	0.153

3. Use case

D. common



- Autonomous Driving
- Robot Navigation
- Drone

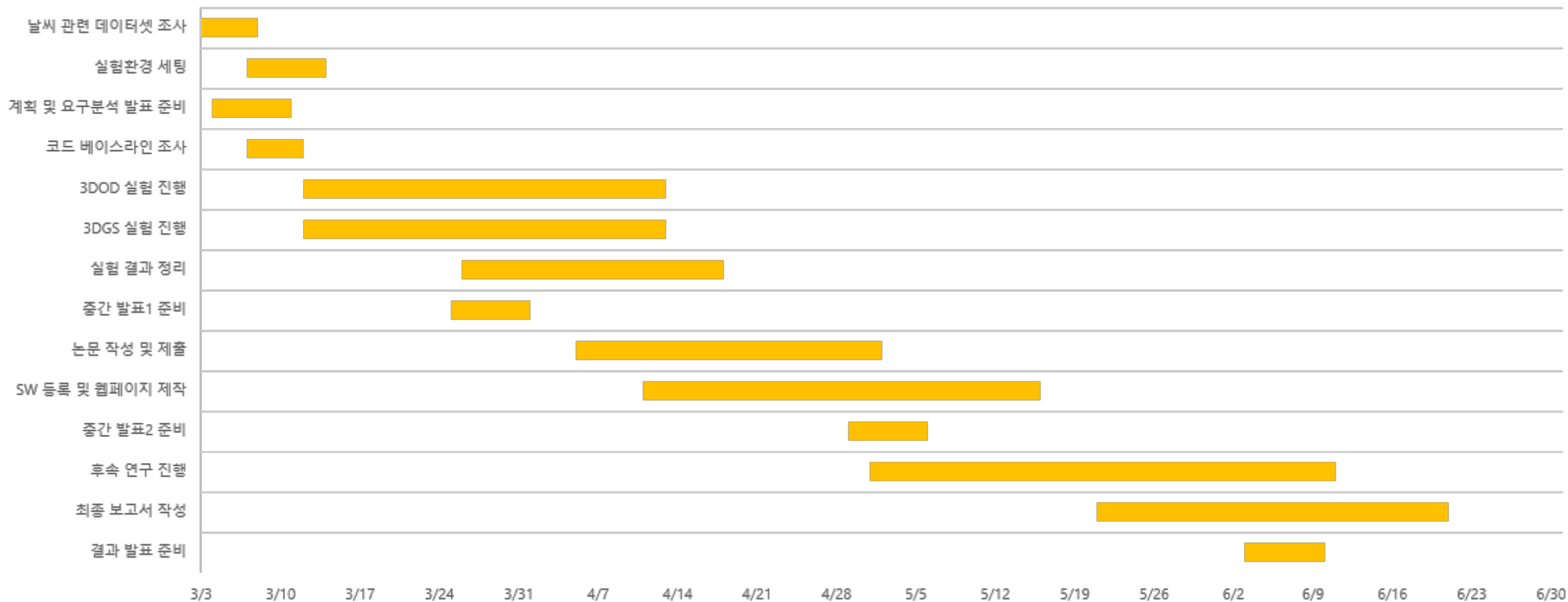
- Digital Twin
- Map Update
- Facility Inspection

- Pre-launch Validation

공동 연구자: 배채은님

4. Future and Schedule

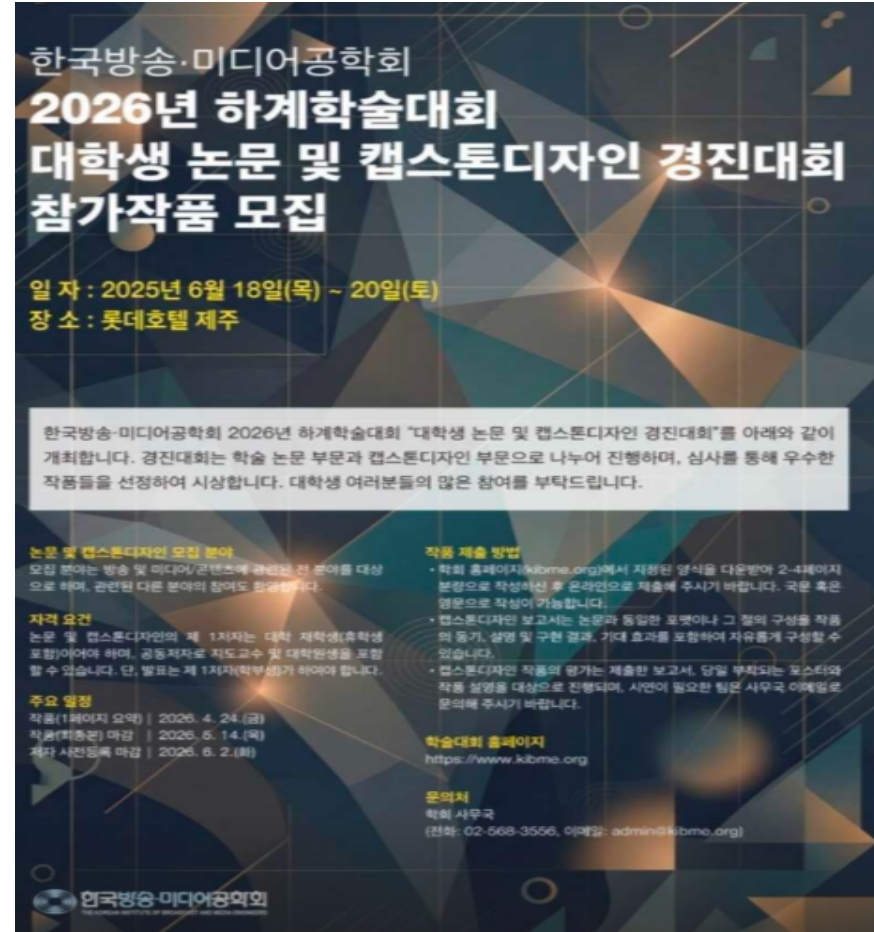
D. common



4. Future and Schedule

D. common

- Paper Writing
- Attending a conference
- Software Registration
- Webpage Creation



한국방송·미디어공학회
2026년 하계학술대회
대학생 논문 및 캡스톤디자인 경진대회
참가작품 모집

일 자 : 2025년 6월 18일(목) ~ 20일(토)
장 소 : 롯데호텔 제주

한국방송·미디어공학회 2026년 하계학술대회 "대학생 논문 및 캡스톤디자인 경진대회"를 아래와 같이 개최합니다. 경진대회는 학술 논문 부문과 캡스톤디자인 부문으로 나누어 진행하며, 심사를 통해 우수한 작품들을 선정하여 시상합니다. 대학생 여러분들의 많은 참여를 부탁드립니다.

논문 및 캡스톤디자인 모집 분야
모집 분야는 방송 및 미디어/콘텐츠에 관련된 전 분야를 대상으로 하며, 관련된 다른 분야의 참여도 가능합니다.

자격 요건
논문 및 캡스톤디자인의 제 1저자는 대학 재학생(휴학생 포함)이어야 하며, 공동저자로 지도교수 및 대학원생을 포함할 수 있습니다. 단, 발표는 제 1저자(학부생)가 하여야 합니다.

주요 일정
작품(1페이지 요약) | 2026. 4. 24.(금)
작품(최종본) 마감 | 2026. 5. 14.(목)
제자 사전등록 마감 | 2026. 6. 2.(화)

작품 제출 방법
· 학회 홈페이지(kibme.org)에서 지정된 양식을 다운로드 2-4페이지 분량으로 작성하신 후 온라인으로 제출해 주시기 바랍니다. 국문 혹은 영문으로 작성이 가능합니다.
· 캡스톤디자인 보고서는 논문과 동일한 포맷이나 그 절의 구성을 작품의 동기, 설명 및 구현 결과, 기대 효과를 포함하여 자유롭게 구성할 수 있습니다.
· 캡스톤디자인 작품의 평가는 제출한 보고서, 당일 부착되는 포스터와 작품 설명을 대상으로 진행되며, 사안이 필요한 경우 사무국 이메일로 문의해 주시기 바랍니다.

학술대회 홈페이지
<https://www.kibme.org>

문의처
학회 사무국
(전화: 02-568-3556, 이메일: admin@kibme.org)

한국방송·미디어공학회
Korea Institute of Broadcasting & Media Engineering



Thank you

Thank you for listening to my presentaton

종합설계 프로젝트2 - 3팀 발표자: 김규도

KYUNGPOOK NATIONAL UNIVERSITY